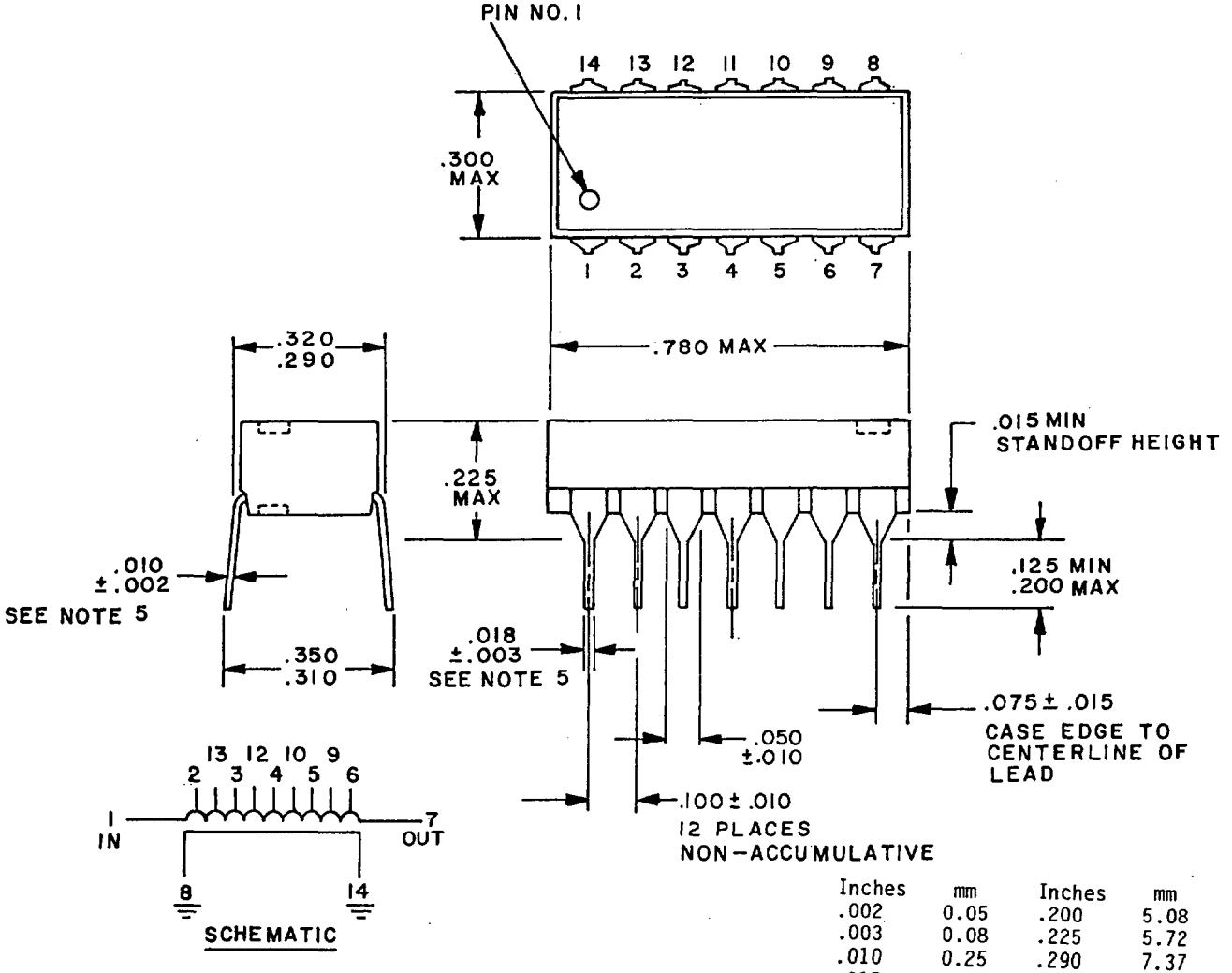
MILITARY SPECIFICATION SHEET

DELAY LINES, 10-TAP, DUAL-IN-LINE, 14-PIN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the delay lines described herein shall consist of this specification and the latest issue of MIL-D-83531.



Inches	mm	Inches	mm
.002	0.05	.200	5.08
.003	0.08	.225	5.72
.010	0.25	.290	7.37
.015	0.38	.300	7.62
.018	0.46	.310	7.87
.050	1.27	.320	8.13
.075	1.91	.350	8.89
.125	3.18		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Location and shape of standoffs are optional. Standoff height is required.
4. Pin 11 is not connected.
5. Terminal dimensions are prior to hot solder dip.
6. Schematic is for general information only.

FIGURE 1. Case and mounting dimensions.

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REQUIREMENTS:

Operating temperature: -55°C to +125°C.

Weight: 2.5 grams maximum.

Case material: Plastic.

Terminal material: Type LR(0.52) G-35-C* in accordance with MIL-STD-1276. Prior to shipment, terminals shall be hot solder dipped in SN60, SN62, or SN63 in accordance with QQ-S-571 to bottom of standoffs, minimum.

Design and construction: See figure 1.

Total delay time: See table I.

Rise time: 25% of total delay, maximum.

Voltage attenuation: See table I.

Distortion: ±10% at maximum delay.

Delay time variation with temperature: 100 ppm/°C maximum.

DC resistance (RDC): $RDC = \text{impedance}\left(\frac{\text{attenuation}}{1 - \text{attenuation}}\right)$

EXAMPLE: $RDC = 50\left(\frac{0.03}{1-0.03}\right)$ RDC = 1.5 ohms

Nominal characteristic impedance: See table I.

Insulation resistance: In accordance with MIL-D-83531.

Input pulse:

Peak pulse voltage: 1 to 3 volts ±5%.

Pulse width: Three times nominal total delay.

Duty cycle: 25%.

Salt spray (corrosion): In accordance with MIL-D-83531.

Thermal shock: In accordance with MIL-D-83531.

Shock: In accordance with MIL-D-83531.

Tap accuracy and total accuracy:

Total delay 0 to 10: ±0.5 ns.
Total delay 11 to 25: ±1 ns or 5% (whichever is greater).
Total delay 26 and up: ±2 ns or 5% (whichever is greater).

Part number: The part number shall be as follows:

EXAMPLE:

M	83531/01	-001
Military designator	Specification sheet number (note zero)	Dash number (see table I)

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TABLE I. Part numbers and electrical characteristics.

Part number M83531/01- Nominal characteristic impedance (±10%)					1/ Total delay time (ns) ±5%	Maximum attenuation by impedance (percent)				
50Ω	100Ω	200Ω	360Ω	500Ω		50Ω	100Ω	200Ω	360Ω	500Ω
001	101	201	301	401	5	3	5	5	5	7
002	102	202	302	402	10	"	"	"	"	"
003	103	203	303	403	15	"	"	"	"	"
004	104	204	304	404	20	"	"	"	"	"
005	105	205	305	405	25	"	"	"	"	"
006	106	206	306	406	30	"	"	"	"	"
007	107	207	307	407	35	"	"	"	"	"
008	108	208	308	408	40	"	"	"	"	"
009	109	209	309	409	45	"	"	"	"	"
010	110	210	310	410	50	"	"	"	"	"
011	111	211	311	411	55	"	"	"	"	"
012	112	212	312	412	60	"	"	"	"	"
013	113	213	313	413	65	"	"	"	"	"
014	114	214	314	414	70	"	"	"	"	"
015	115	215	315	415	75	"	"	"	"	"
016	116	216	316	416	80	"	"	"	"	"
017	117	217	317	417	85	"	"	"	"	"
018	118	218	318	418	90	"	"	"	"	"
019	119	219	319	419	95	"	"	"	"	"
020	120	220	320	420	100	"	"	"	"	"
021	121	221	321	421	110	"	"	6	7	8
022	122	222	322	422	120	"	"	"	"	"
023	123	223	323	423	130	"	"	"	"	"
024	124	224	324	424	140	"	"	"	"	"
025	125	225	325	425	150	"	"	"	"	"
026	126	226	326		160	"	"	"	"	"
027	127	227	327		170	"	"	"	"	"
028	128	228	328		180	"	"	"	"	"
029	129	229	329		190	"	"	"	"	"
030	130	230	330		200	5	"	"	"	"
031	131	231	331		210	"	"	"	"	"
032	132	232	332		220	"	"	"	"	"
033	133	233			230	"	"	"	"	"
034	134	234			240	"	"	"	"	"
035	135	235			250	"	"	"	"	"
036	136	236			260	"	"	"	"	"
037	137	237			270	"	"	"	"	"
038	138	238			280	"	"	"	"	"
039	139	239			290	"	"	"	"	"
040	140	240			300	"	"	"	"	"
041	141	241			310	"	7	8	10	10
042	142	242			320	"	"	"	"	"
043	143	243			330	"	"	"	"	"
044	144	244			340	"	"	"	"	"
045	145	245			350	"	"	"	"	"
046	146	246			360	"	"	"	"	"
047	147	247			370	"	"	"	"	"
048	148	248			380	"	"	"	"	"
049	149	249			390	"	"	"	"	"
050	150	250			400	"	"	"	"	"
051	151				410	"	8	10	"	"
052	152				420	"	"	"	"	"
053	153				430	"	"	"	"	"
054	154				440	"	"	"	"	"
055	155				450	"	"	"	"	"
056	156				460	"	"	"	"	"
057	157				470	"	"	"	"	"
058	158				480	"	"	"	"	"
059	159				490	"	"	"	"	"
060	160				500	"	"	"	"	"

1/ Delay between successive taps shall be 1/10 of the nominal total delay time.

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QUALITY ASSURANCE PROVISIONS:

Extent of qualification: The extent of qualification shall be as specified in MIL-D-83531, for example:

> Qualification and testing of M83531/01-001 and M83531/01-425 shall be sufficient to grant qualification approval to M83531/01-001 through M83531/01-425 inclusive.

Supersession data: MIL-D-83531/1 supersedes MIL-D-23859/23 when qualified sources are available.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5999-0170-01)

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